

# Transformations and Decompositions

Monday, 12 February 2024 10:46

| Lecture 1                                                                                                                                                                                                                             | Lecture 2                                                                                                                                                                                                                                     | Lecture 3                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Data Structures, Vectors, and Matrices</li> <li>Vector Operations and Vector Spaces</li> <li>Vector Norms</li> <li>K-Nearest Neighbors</li> <li>Dot Products</li> <li>Linear Models</li> </ul> | <ul style="list-style-type: none"> <li>Matrix Multiplication</li> <li>Ordinary Least Squares (OLS)</li> <li>Matrix Transformations</li> <li>Basis, Change of Basis</li> <li>Eigen Algebra</li> <li>Principal Components Regression</li> </ul> | <ul style="list-style-type: none"> <li>Hypothesis Testing</li> <li>Random Variables</li> <li>Probability Distributions (discrete, continuous)</li> </ul> |

## Prediction Problem

- Let's revisit the regression problem from last lecture
- The graph of Spend in Category vs Checkout Amount might look like:



| Visitor ID | Product Category | Spend in Category | Prime member | Purchased products | Checkout amount |
|------------|------------------|-------------------|--------------|--------------------|-----------------|
| 84920594   | AB044391         | 0                 | 0            | 0                  | 0               |
| 84225954   | AB044391         | 70                | 42           | 1                  | 156             |
| 83202384   | AB044391         | 30                | 15           | 0                  | 0               |
| 84342229   | AB044391         | 45                | 12           | 1                  | 79              |
| 8559532    | AB044391         | 0                 | 43           | 0                  | 0               |
| 84332252   | AB044391         | 0                 | 0            | 0                  | 0               |
| 84453002   | AB044391         | 68                | 35           | 1                  | 222             |
| 84555756   | AB044391         | 0                 | 40           | 0                  | 0               |
| 8455452    | AB044391         | 0                 | 32           | 0                  | 0               |
| 8333443    | AB044391         | 0                 | 0            | 1                  | 44              |
| 84450232   | AB044391         | 0                 | 12           | 0                  | 0               |
| 48886763   | AB044391         | 0                 | 46           | 1                  | 234             |
| 85538675   | AB044391         | 57                | 55           | 1                  | 255             |
| 84090532   | AB044391         | 0                 | 24           | 0                  | 0               |
| 84228525   | AB044391         | 75                | 41           | 1                  | 204             |
| 83454321   | AB044391         | 0                 | 23           | 0                  | 0               |
| 84567349   | AB044391         | 0                 | 8            | 1                  | 15              |

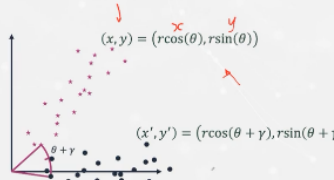
## Prediction Problem

- Instead of applying the KNN algorithm to make predictions, find the (linear) relationship



## Rotating Data

- One way to do this is by using angles to rotate the data



$(x, y) = (r \cos(\theta), r \sin(\theta))$   
 $(x', y') = (r \cos(\theta + \gamma), r \sin(\theta + \gamma))$

## Rotating Data

- One way to do this is by using angles to rotate the data

$(x, y) = (r \cos(\theta), r \sin(\theta))$   
 $\downarrow$   
 $(x', y') = (r \cos(\theta + \gamma), r \sin(\theta + \gamma))$   
 $= (r \cos(\theta) \cos(\gamma) - r \sin(\theta) \sin(\gamma), r \sin(\theta) \cos(\gamma) + r \cos(\theta) \sin(\gamma))$



## Rotating Data

- One way to do this is by using angles to rotate the data

$(x', y') = (r \cos(\theta + \gamma), r \sin(\theta + \gamma))$   
 $= (x \cos(\gamma) - y \sin(\gamma), y \cos(\gamma) + x \sin(\gamma))$   
 $= \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) \\ \sin(\gamma) & \cos(\gamma) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$



The fact of having rotated the axes by adding an angle allows us to find the linear relationship entre variables (no simple as evident as here where if one value grows the other grows in almost the same proportion using the product between matrices and the sine and cosine of the added angle to achieve the rotation on the coordinate axis

## MATRIX MULTIPLICATION

To perform the final calculation, we need to know how to multiply matrices with different dimensions

## Matrix Multiplication

- Matrix product definition

$$[AB]_{i,j} = \sum_l a_{i,l} b_{l,j}$$

$\begin{bmatrix} a_{11} & \dots & a_{1k} \end{bmatrix} \begin{bmatrix} b_{11} & \dots & b_{1m} \end{bmatrix}$

- Dot Products of columns/rows

To do the Dot Product, the number of rows of one must equal the number of columns of the other commutative since unless they have the same dimension, changing order won't match rows with columns

$$P = AB = \begin{bmatrix} \vdots & \vdots & \vdots \\ a_{n1} & \dots & a_{nk} \end{bmatrix} \begin{bmatrix} \vdots & \vdots & \vdots \\ b_{k1} & \dots & b_{km} \end{bmatrix}$$

$[n \times k] \quad [k \times m]$

To do the Dot Product, the number of rows of one must equal the number of columns of the other commutative since unless they have the same dimension, changing order won't match rows with columns

## Matrix Multiplication

### Matrix product example

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 \\ 5 & 1 \end{bmatrix}$$

$$P = AB = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \times 1 + 3 \times 5 & 2 \times 4 + 3 \times 1 \\ 0 \times 1 + 2 \times 5 & 0 \times 4 + 2 \times 1 \end{bmatrix}$$

## Matrix Multiplication

### Properties of Matrix Multiplication

#### Distributivity

$$A(B + C) = AB + AC$$

#### Associativity

$$A(BC) = (AB)C$$

#### Not Commutativity

$$AB \neq BA$$

## MATRIX FACTORIZATION OR DECOMPOSITION

### Factorization

- Factorization is multiplication in reverse
  - » Break a mathematical object into smaller parts
  - » Combine parts with an operator to get back original object
- Factor the number 30
  - » Find numbers when multiplied together, equal 30
  - » 2 & 15
  - » 6 & 5
  - » 2 & 3 & 5

Given an original matrix, find the matrices that when multiplied give that original matrix as a result

This has a great impact on operations with matrices since with millions of rows and columns we can greatly reduce computation time

## Matrix Factorization

- Use a matrix as the mathematical object
  - » Start with a matrix
  - » Find other matrices that equal the original when multiplied

"Many key ideas of linear algebra, when you look at them closely, are really factorizations of a matrix. The original matrix becomes the product of 2 or 3 special matrices." - Gilbert Strang

different operations to factorize a matrix

## Matrix Factorization

- Popular matrix factorizations
  - » Matrix factorization can reduce the dimensionality of an ML problem (dataset)
  - »  $A = LU$  = (lower triangular  $L$ ) (upper triangular  $U$ )
  - »  $A = QR$  = (orthogonal columns in  $Q$ ) (upper triangular  $R$ )
  - »  $A = Q\Lambda Q^T$  = (eigenvectors in  $Q$ ) (eigenvalues in  $\Lambda$ )
  - »  $A = U\Sigma V^T$  = (singular vectors in  $U$  and  $V$ ) (singular values in  $\Sigma$ )
  - »  $A = CR$  = (describes the columns space  $C$ ) (describes the rows space  $R$ )

ColumnRow Factorization (CR) Finding columns

## CR Matrix Factorization

### Solving for $C \& R$

- Start by treating each column as its own vector

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} x_2 + \begin{bmatrix} 5 \\ 5 \\ 3 \end{bmatrix} x_3$$

## CR Matrix Factorization

### Solving for $C \& R$

#### Independent

A set of vectors that cannot be equal with any scalar multiplier nor addition of other vectors from the matrix

$$\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$

$\vec{v}_1 \quad \vec{v}_2$

#### Dependent

A set of vectors that can be constructed by scalar multipliers and/or addition of other vectors from the matrix

$$\begin{bmatrix} 5 \\ 5 \\ 3 \end{bmatrix}$$

$\vec{v}_3$

### Solving for $C \& R$

- Start by treating each column as its own vector

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} x_2 + \begin{bmatrix} 5 \\ 5 \\ 3 \end{bmatrix} x_3$$

$$\vec{v}_1 + \vec{v}_2 = \vec{v}_3$$

- Look for columns that can be scaled and added to get other columns
  - » Columns that depend on the other columns

This is an evident example that the 3rd column can be calculated

To simplify the problem, we see that if we try to calculate the dependent vector with the other two independent vectors simply by adding them we get the exact result the following exercise is to FACTORIZE the matrix to find which matrix multiplied by the independent variables matrix gives us the original matrix

## CR Matrix Factorization

### Solving for $C \& R$

### Solving for $C \& R$

» Start with an independent set of vectors to reconstruct  $A$   
 » Then look for a second matrix that will result in  $A$

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \end{bmatrix} = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

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» Start with an independent set of vectors to reconstruct  $A$   
 » Then look for a second matrix that will result in  $A$

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \end{bmatrix} = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\begin{cases} 1 \times R_{11} + 4 \times R_{21} = 1 \\ 3 \times R_{11} + 2 \times R_{21} = 3 \\ 2 \times R_{11} + 1 \times R_{21} = 2 \end{cases}$$

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With 3 equations and 2 unknowns we can solve it

• Solving for  $C$  &  $R$

» Start with an independent set of vectors to reconstruct  $A$   
 » Then look for a second matrix that will result in  $A$

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \end{bmatrix} = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\begin{cases} 1 \times 1 + 4 \times 0 = 1 \\ 3 \times 1 + 2 \times 0 = 3 \\ 2 \times 1 + 1 \times 0 = 2 \end{cases}$$

$$\begin{cases} R_{11} = 1 \\ R_{21} = 0 \end{cases}$$

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• Solving for  $C$  &  $R$

$$C = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

» The two matrices,  $C$  &  $R$ , equal  $A$  when multiplied together

$$A = CR = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

## CR Matrix Factorization

- These matrices form a foundation for reconstructing matrix  $A$

$$C = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad A = CR = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

- $C$  &  $R$  form a **basis**

- $C$  is a Basis of the **Column Space**
- $R$  is a Basis of the **Row Space**
- They are the minimal set of vectors required to get back to the original matrix
- Number of columns in  $C$  must equal the number of rows in  $R$

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## CR Matrix Factorization

- These matrices form a foundation for reconstructing matrix  $A$

$$C = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad A = CR = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

- Rank**

- Number of independent columns/rows in a matrix
- The  $CR$  factorization is **rank** revealing:
  - The smallest number of columns for which the product  $CR$  reproduces  $A$
- For very large matrices... this process is NOT easy!

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## MATRIX TRANSFORMATIONS

We can view matrices as a set of equations and variable values  
 To solve some matrix operations, like with vectors, we can use geometria.

## Matrix Transformations

- Matrices are more than just a way to store data, they map/transform data
  - Matrix multiplication skews, rotates, and scales vectors
  - Can transformations be **reversed**?

$$\begin{bmatrix} \cos y & -\sin y \\ \sin y & \cos y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

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The transpose matrix results from swapping rows for  
 It can be useful for operations  
 dependent on the matrix dimension  
 equals its transpose is called a symmetric matrix

$$A_{ij} \quad AT = A_{j,i}$$

The identity matrix is the one that when multiplied by the  
 original matrix does not alter the result

$$A \times I = A$$

The inverse matrix is the one that when multiplied by the  
 original gives us the identity matrix

$$A \times A^{-1} = I$$

The inverse matrix has all 1s on the diagonal and 0s elsewhere

$$\begin{bmatrix} 1 & 4 & 0 \\ 3 & 2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

symmetric

## Matrix Inverse

- Reminder: The reciprocal is a number/function/set of functions that returns the identity

$$8 \cdot \frac{1}{8} = 1$$

- The number 1 is the identity because  $8 \cdot 1 = 8$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} = I$$

$$AI = A$$

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- Reminder: The reciprocal is a number/function/set of functions that returns the identity

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} = I$$

$$AI = A$$

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- Reminder: The reciprocal is a number/function/set of functions that returns the identity

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Applying the inverse matrix is like changing the axis of

Applying the inverse matrix is like changing the axis of

### Matrix Inverse

- Inverse

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

El determinante de una matriz es el área que lo contiene

- How to find the inverse of a matrix

$A_1 = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$

$\text{Det}(A_1) = \text{Area} = a \cdot d$

- Inverse

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

- How to find the inverse of a matrix

$A_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\text{Area}_{\text{square}} = (a+b)(c+d)$

- How to find the inverse of a matrix

$A_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\text{Det}(A_2) = \text{Area}_{\text{parallelogram}} = (a+b)(c+d) - b \cdot c - b \cdot c - a \cdot c - b \cdot d$

The denominator cannot be 0

- How to find the inverse of a matrix

$A_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\text{Det}(A_2) = \text{Area}_{\text{parallelogram}} = (a+b)(c+d) - b \cdot c - b \cdot c - a \cdot c - b \cdot d$

$= ac + ad + bc + bd - 2bc - ac - bd$

$= ad - bc$

We'll use the matrix determinant to know if the inverse matrix can be applied

- The matrix inverse of a 2D linear transformation is given by

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\text{det}(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

- If the **determinant of a matrix is zero**, the matrix does not have an inverse
- The linear transformation cannot be undone

The inverse of a matrix, and therefore the possibility of reverting a linear transformation, depends on the determinant value of that matrix

If the determinant of a matrix is zero, it means the matrix has no inverse, in other words, it is and the linear transformation cannot deshacer.

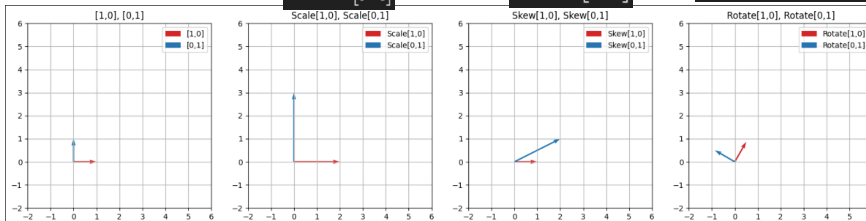
Other operations we can perform are scale, skew, and rotate

Original Matrix

Scale =  $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

Skew =  $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Rotate =  $\begin{bmatrix} \cos(\pi/3) & -\sin(\pi/3) \\ \sin(\pi/3) & \cos(\pi/3) \end{bmatrix}$



Besides visualizing operation results, we can examine not only how individual vectors change, but also

A quick calculation reveals that scaling changes the area but skew and rotation do not affect the transformed area to what extent an area in original space is scaled by the transformation lineal.

An analytical solution versus a

It's about finding an analytical solution operating on matrices that allow us to estimate y values based on the linear dependence equation

differential calculus solution seeks the best result of the estimation by ANALYZING the result

### Prediction Problem

- Find the relationship in the data by rotating the data, or rotating the x-axis so it goes through the middle of the data

Now we want to know not if they buy but how much they will spend

We could compare results with KNN Regressor

- Find the rotation that minimizes the overall orthogonal distance to the new x-axis
- How?
  - Could use gradient descent
  - But for this linear problem, an exact solution can be constructed!

The first assumption is that there is a between the feature variable values model feature (Checkout quantity) que queremos predecir, y por tanto it can be expressed as a weighted sum of features

solving linear regression reduces to finding a line that best

In a more general situation with more regressor features best fits the multidimensional dataset

The method to find the line is not as rigorous as MLE which requires differential calculus

While simple problems like linear regressions can admit analytical solutions like OLS quite restrictive in finding a solution

$\text{Checkout Amount} = b + \text{Spent in Category} \cdot w_{\text{Spent in Category}}$

$\mathbf{w} = [b, w_{\text{Spent in Category}}]$

$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}$

$\mathbf{y} = \mathbf{X}\mathbf{w}$

$\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$

Cuando X no es una matriz cuadrada, podemos transponerla para que si lo sea

$\mathbf{X}^T\mathbf{y} = (\mathbf{X}^T\mathbf{X})\mathbf{w}$

La Matriz cuadrada resultante es la matriz de Covarianza (dimensional scaler) y tiene una inversa

$\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

Si usamos nuestro dataset e intentamos comparar el resultado frente a KNN regresor, vemos que el resultado es peor, posiblemente porque una sola línea es un modelo demasiado sencillo para capturar la no linealidad de nuestro dataset. Una manera de arreglarlo podría ser añadiendo no linealidad al modelo con features adicionales. El problema de ir añadiendo variables a OLS es que empieza a ser computacionalmente costoso, y no tenemos garantías de que se consigan resultados.

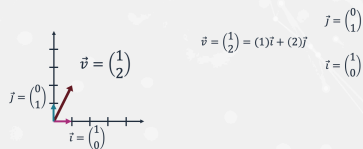
## CAMBIO DE BASE Y VECTORES EIGEN

To remove the OLS restriction where  $\det(A)$  cannot be 0  
continue making transformations to solve it

An important PCA transformation is Change of Basis, which is rotating the coordinate coordinates

### Change of Basis

- Start with a vector

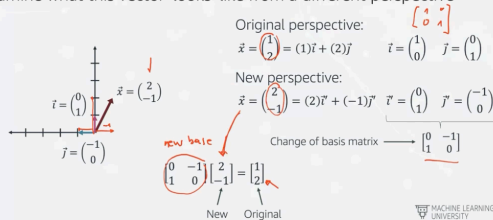


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An interesting thing is that multiplying the new basis by the new result gives us the  
Mathematically this is how you change the basis

### Change of Basis

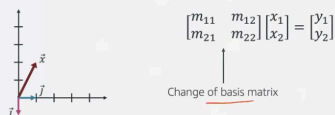
- Examine what this vector 'looks' like from a different perspective



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### Change of Basis

- Change of basis matrices & transformations can be combined
  - Express transformations on various vectors and bases
- Alternate coordinate systems are important
  - Lead to matrices with inverses or easy to compute inverses



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### Eigen Algebra

- Eigenvectors** are a special case of vectors
  - The **direction** after being multiplied by a matrix **does not change**
  - The vector is scaled (stretched or shrunk)
- Eigenvalues**
  - Measure **how much** an eigenvector gets stretched or shrunk

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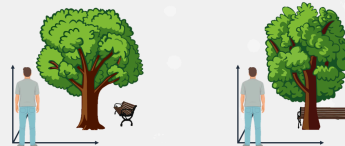
### Eigen Algebra

- Mathematically represent the movement around the tree
  - Branches, roots, and bench are all vectors that change their directions
  - The trunk of the tree would keep its place, its direction



### Basis

- This is how things work in the real world
  - Objects start in space, then you put a coordinate system on top of the objects
  - The coordinates can have any perspective, just like the vector before



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These basis transformations

Lead to using Eigen Algebra

Allow us to easily compute inverse matrices when it wasn't possible

We will use another matrix factorization method we haven't used:  
EIGEN factorization

- Finding eigenvectors and eigenvalues

- When  $(A - \lambda I)$  is not invertible

$$\text{Det}(A - \lambda I) = ad - bc - a\lambda - d\lambda + \lambda^2$$

- Now there is a quadratic to solve and use with:

$$A\vec{v} = \lambda\vec{v}$$





## Eigen Algebra

### What is meant by Eigenvector?

- » Start with a vector  $\vec{v}$
- » Multiply it by matrix  $A$

$$A\vec{v}$$

$$A\vec{v} = \lambda\vec{v}$$

- » Results in the same vector, but scaled

### What is meant by Eigenvector?

- » Results in the same vector, but scaled

$$A\vec{v} = \lambda\vec{v}$$

- » Meaning:

- » Both  $A\vec{v}$  and  $\vec{v}$  live on the line spanned by  $\vec{v}$
- » The resulting vector is pointing in the same (or opposite) direction after matrix multiplication

## Eigen Algebra

### Example

- » Is  $\vec{v}$  an eigenvector of  $A$ ?

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A\vec{v} =$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

## Eigen Algebra

### Example

- » What's the eigenvalue of  $\vec{v}$ ?

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A\vec{v} = 2\vec{v}$$

$$\lambda = 2$$

## Eigen Algebra

### Finding eigenvectors and eigenvalues

- » Eigenvector
  - » A vector that maintains its original span after a linear transformation
- » Eigenvalue
  - » The value by which an eigenvector is scaled
- » Solve this equation

$$A\vec{v} = \lambda\vec{v}$$

## PRINCIPAL COMPONENT ANALYSIS

## Eigen Algebra

### Eigenvectors and Principal Components

- » Goal: find a basis that will transform the data set along a direction that best fits the data → that is the principal component
- » Steps:
  - » Center the data
  - » Compute the covariance matrix
  - » Find eigenvectors of the covariance matrix
  - » The technique of extracting the principal components of a data set is known as **Principal Component Analysis (PCA)**
  - » Regression on a data set created with PCA is called **Principal Components Regression (PCR)**

Once we obtain the principal component with PCA, we can Perform a regression on it with PCR choosing a regression method OLS regressor, etc).

### Eigenvectors and Principal Components

- » First center the data by subtracting the mean in each direction



### Eigenvectors and Principal Components

- » Multiply the data by its transpose to get the Covariance matrix

$$A^T A = \begin{bmatrix} 0 & 70 & 30 & 45 & 0 & 0 & 68 & 0 & 0 & 0 & 0 & 0 \\ 0 & 42 & 15 & 12 & 43 & 0 & 35 & 40 & 32 & 0 & 12 & 46 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

### Eigenvectors and Principal Components

- » Multiply the data by its transpose to get the Covariance matrix

$$A^T A = \begin{bmatrix} \text{VAR}(A_{11}A_{12}) & \text{COV}(A_{11}A_{12}) & \dots & \text{COV}(A_{11}A_{1n-1}) & \text{COV}(A_{11}A_{1n}) \\ \text{COV}(A_{21}A_{11}) & \text{VAR}(A_{21}A_{12}) & \dots & \text{COV}(A_{21}A_{1n-1}) & \text{COV}(A_{21}A_{1n}) \\ \vdots & \vdots & \ddots & \vdots & \vdots \end{bmatrix}$$

### Eigenvectors and Principal Components

- » The Eigenvector with the **maximum** Eigenvalue of the covariance matrix, after centering the data, is the variable (direction) that **best** represents the data

$$\text{Prime} \rightarrow \vec{v} \rightarrow \text{Max}(\lambda)$$

$$\begin{bmatrix} \text{COV}(A_{n-11}A_{11}) & \text{COV}(A_{n-11}A_{12}) & \dots & \text{VAR}(A_{n-11}A_{1n-1}) & \text{COV}(A_{n-11}A_{1n}) \\ \text{COV}(A_{n1}A_{11}) & \text{COV}(A_{n1}A_{12}) & \dots & \text{COV}(A_{n1}A_{1n-1}) & \text{VAR}(A_{n1}A_{1n}) \end{bmatrix}$$

- » The Eigenvalues of this matrix represent the amount of variance (or stretch) in a given direction
- » Select the direction with the Maximum Eigenvalue



There is another PCA factorization method if you don't want EIGEN vectors:

Single Value Decomposition (SVD) to solve with OLS when the inverse. Eigen opera sólo sobre matrices cuadradas, cuando esto no es posible, se puede utilizar la factorización matrices SVD, que opera sobre matrices rectángulares.

#### • Popular matrix factorizations

- » Matrix factorization can reduce the dimensionality of an ML problem (dataset)

- »  $A = LU$  = (lower triangular  $L$ ) (upper triangular  $U$ )
- »  $A = QR$  = (orthogonal columns in  $Q$ ) (upper triangular  $R$ )
- »  $A = Q\Lambda Q^T$  = (eigenvectors in  $Q$ ) (eigenvalues in  $\Lambda$ )
- »  $A = U\Sigma V^T$  = (singular vectors in  $U$  and  $V$ ) (singular values in  $\Sigma$ )
- »  $A = CR$  = (describes the columns space  $C$ ) (describes the rows space  $R$ )

#### What is SVD?

The eigenvalues and eigenvectors are defined only for squared matrices. The Singular Value Decomposition (SVD) is a matrix factorization that generalizes the eigendecomposition of a square normal matrix to **any rectangular matrix**. For any  $n \times m$  matrix  $\mathbf{X}$ , the following decomposition always exists:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

where  $\mathbf{U}$  is a column-orthonormal  $n \times r$  matrix,  $r$  is the rank of the matrix  $\mathbf{X}$  (number of linearly independent rows or columns),  $\mathbf{\Sigma}$  is a diagonal  $r \times r$  matrix where the elements are sorted in descending order, and  $\mathbf{V}$  is a column-orthonormal  $r \times m$  matrix.

The diagonal entries, known as the **singular values** of  $\mathbf{X}$ , are the square roots of the eigenvalues of  $(\mathbf{X}\mathbf{X}^T)$  and  $(\mathbf{X}^T\mathbf{X})$ . The number of non-zero singular values is equal to the rank of  $\mathbf{X}$ . The columns of  $\mathbf{U}$  and the columns of  $\mathbf{V}$ , called the **left-singular vectors** and **right-singular vectors** of  $\mathbf{X}$ , respectively, are the eigenvectors of  $(\mathbf{X}\mathbf{X}^T)$  and  $(\mathbf{X}^T\mathbf{X})$ , respectively. Numpy's `np.linalg.svd()` is a good algorithm to obtain the SVD of a matrix, returning all three components.

In summary, for a given dataset  $\mathbf{X}$ , here are the steps to implement SVD of the original data matrix to extract the change of basis and transform the dataset:

- standardize the  $\mathbf{X}$  features dataset (at least center the dataset by subtracting the mean)
- compute the SVD of the  $\frac{1}{\sqrt{n-1}}\mathbf{X}$  matrix, and use the right-singular vectors to define the change of basis  $\mathbf{P}$
- derive the new dataset  $\mathbf{X}_{\text{new}} = \mathbf{X}\mathbf{P}$ , performing the change of basis
- apply linear regression on the new dataset  $\mathbf{X}_{\text{new}}$

<https://arxiv.org/pdf/1404.1100.pdf>